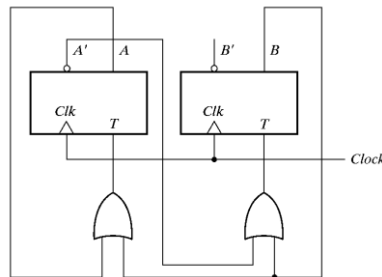


Theoretical Assignment 3 25F

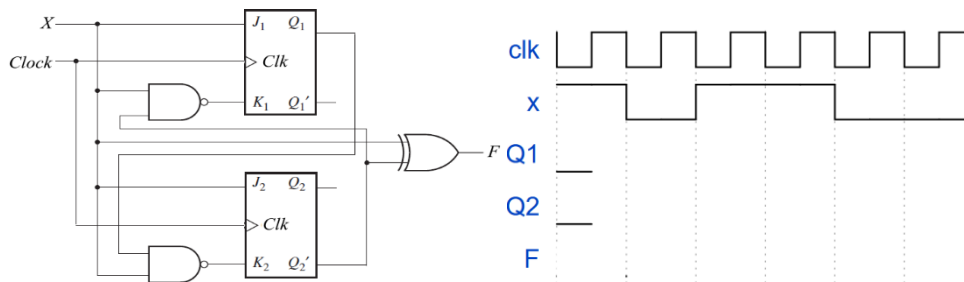
1. A sequential circuit has two JK flip-flops A and B and one input x. The circuit is described by the following flip-flop input equations:

$$J_A = x, K_A = B, J_B = x, K_B = A'$$

- a) Derive the state equations $A(t + 1)$ and $B(t + 1)$ by substituting the input equations for the J and K variables.
 - b) Draw the state table and state diagram of the circuit.
2. Derive the input/state/output function, state table/diagram (**if applicable**), for the sequential circuit shown in the block diagram blow. Explain the function that the circuit performs.



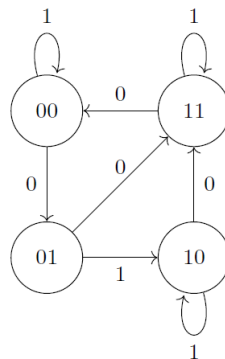
3. For the block diagram below
- a) Derive the input/state/output function, state table/diagram of the sequential circuit if applicable.
 - b) Is it a Mealy machine or a Moore machine? Why?
 - c) Construct a timing diagram for the circuit for an input sequence $X = 101100$. (Assume that initially $Q_1 = Q_2 = 0$, Complete the following waveform for Q_1 , Q_2 and F .)



4. For the following state table

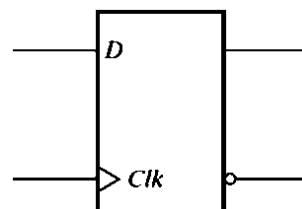
Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

- a) Minimize the state table, show your procedures, and then draw the reduced table and its corresponding state diagram.
- b) Determine the output sequence for input sequence 01010010111 (from left to right) with the original state table and the reduced state table, starting from a. **(Refer to sequence table in Lecture7Page25)**
5. Obtain the simplified input equations for a sequential circuit that uses T flip-flops specified by the state diagram below. **Write down the necessary procedure.** (No need to draw the logic diagram, **Refer to Lecture7Page43: Design Example 3: Design with TFF**)



6. Design a FlipFlop with the function table described below. Apart of the clock input, the FlipFlop has two inputs G and In. **Write down necessary design procedure** as we learnt in class and then draw the block diagram. (Using DFF show in the graphic symbol with extra basic gates).

Clk	G	In	Q	Q'
0	x	x	last Q	last Q'
1	x	x	last Q	last Q'
$\uparrow$	1	0	0	1
$\uparrow$	1	1	1	0
$\uparrow$	0	x	last Q	last Q'



7. Draw the **state diagram** of the machine described by the Verilog model given below.

```

module test (
    output reg y_out,
    input x_in, clk, reset
);
    parameter s0 = 2'b00, s1 = 2'b01, s2 = 2'b10, s3 = 2'b11;
    reg [1:0] state, next_state;

    always @(posedge clk, negedge reset) begin
        if (reset == 1'b0)
            state <= s0;
        else
            state <= next_state;
    end
  
```

```
end

always @(state, x_in) begin
    y_out = 0;
    next_state = s0;
    case (state)
        s0: if (x_in == 1) begin
                y_out = 0;
                if (x_in) next_state = s1;
                else next_state = s0;
            end
        s1: if (x_in == 1) begin
                y_out = 0;
                if (x_in) next_state = s2;
                else next_state = s1;
            end
        s2: if (x_in == 1) begin
                next_state = s3;
                y_out = 0;
            end
            else begin
                next_state = s2;
                y_out = 1;
            end
        s3: if (x_in == 1) begin
                y_out = 1;
                if (x_in) next_state = s0;
                else next_state = s3;
            end
        default: next_state = s0;
    endcase
end
endmodule
```